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Framwork for AI Agent Network

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Framwork for AI Agent Network

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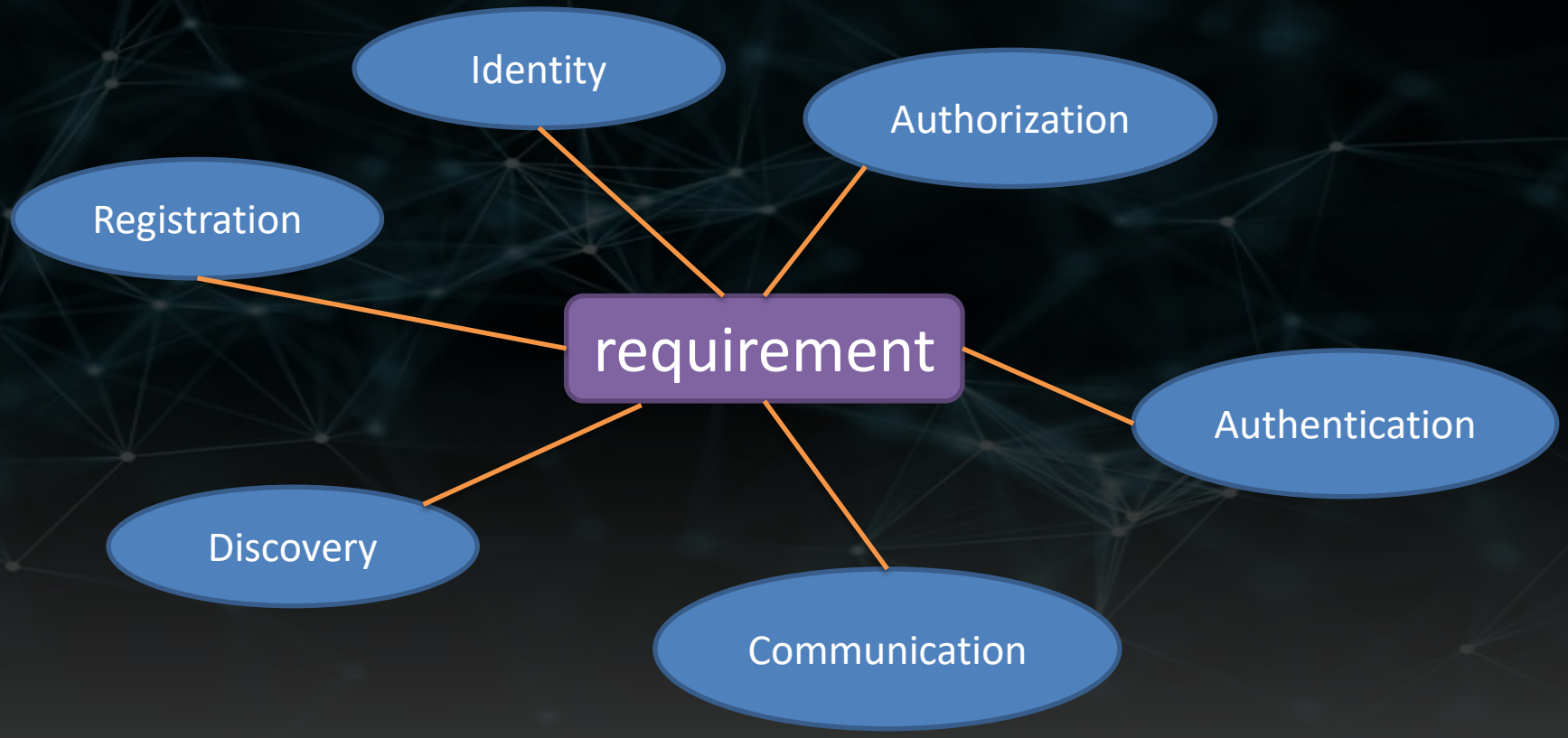

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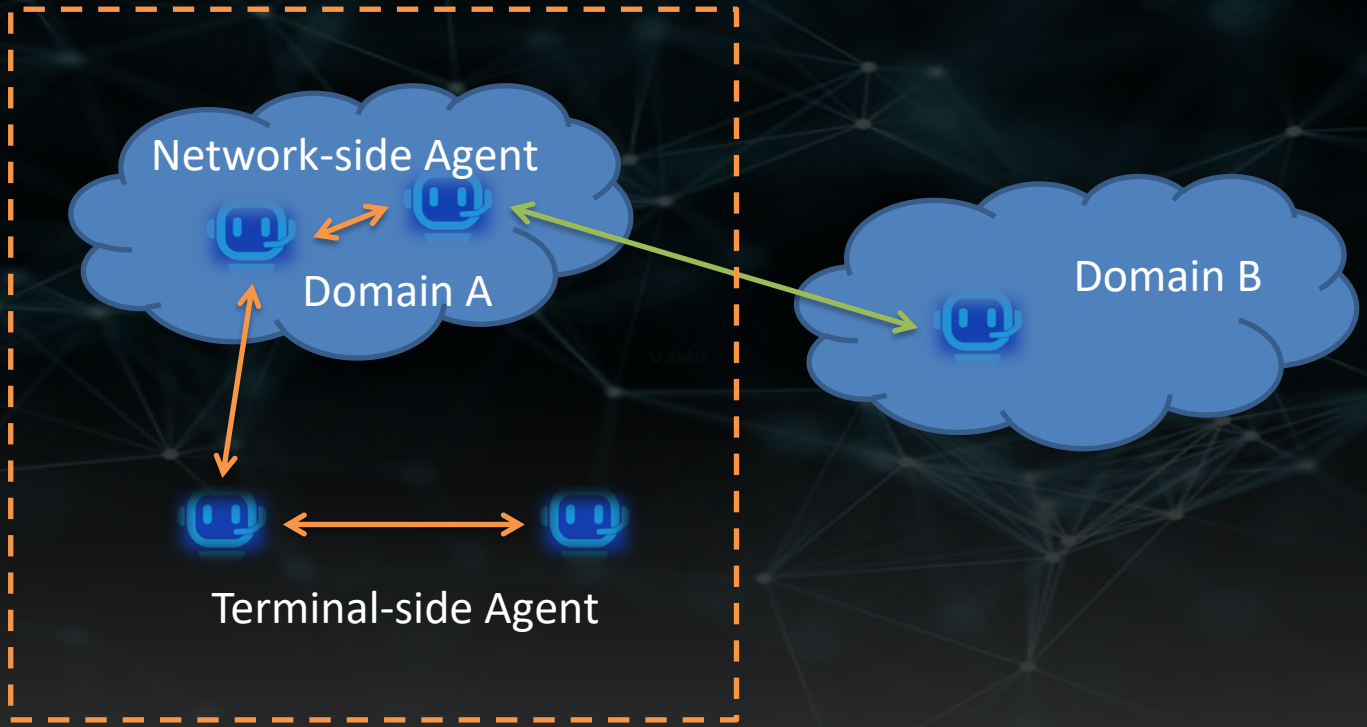

Explanation on this draft

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AI Agent A

Agent Registration Server

Agent Communication Server

AI Agent B

Role

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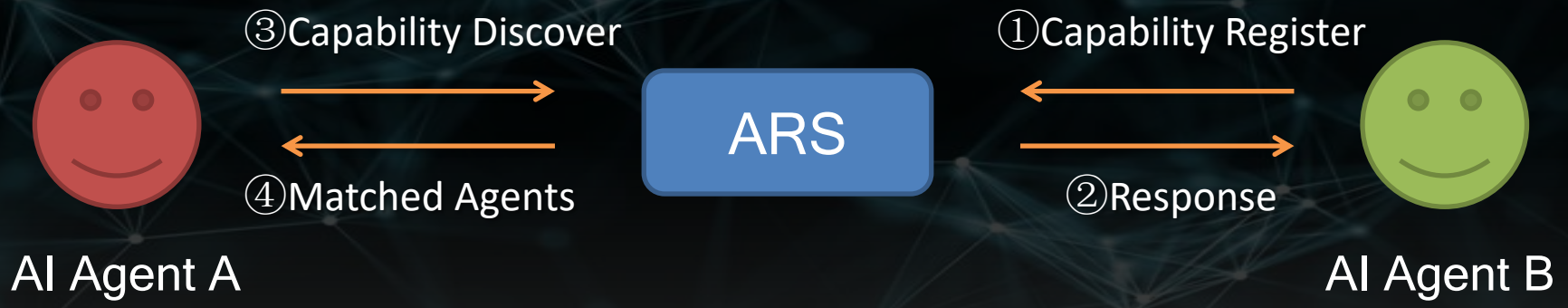

Register and Discover

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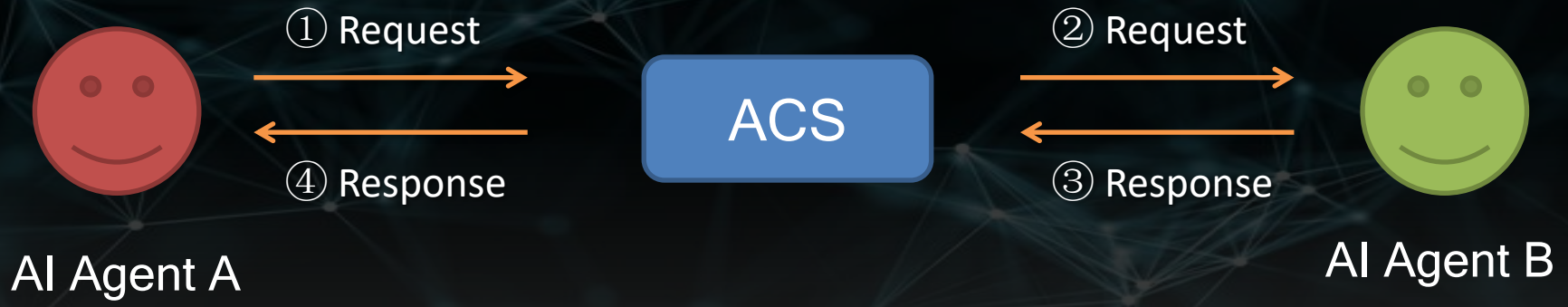

Communication - Option 1

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AI Agent A

② Request

③ Response

AI Agent B

Communication - Option 2

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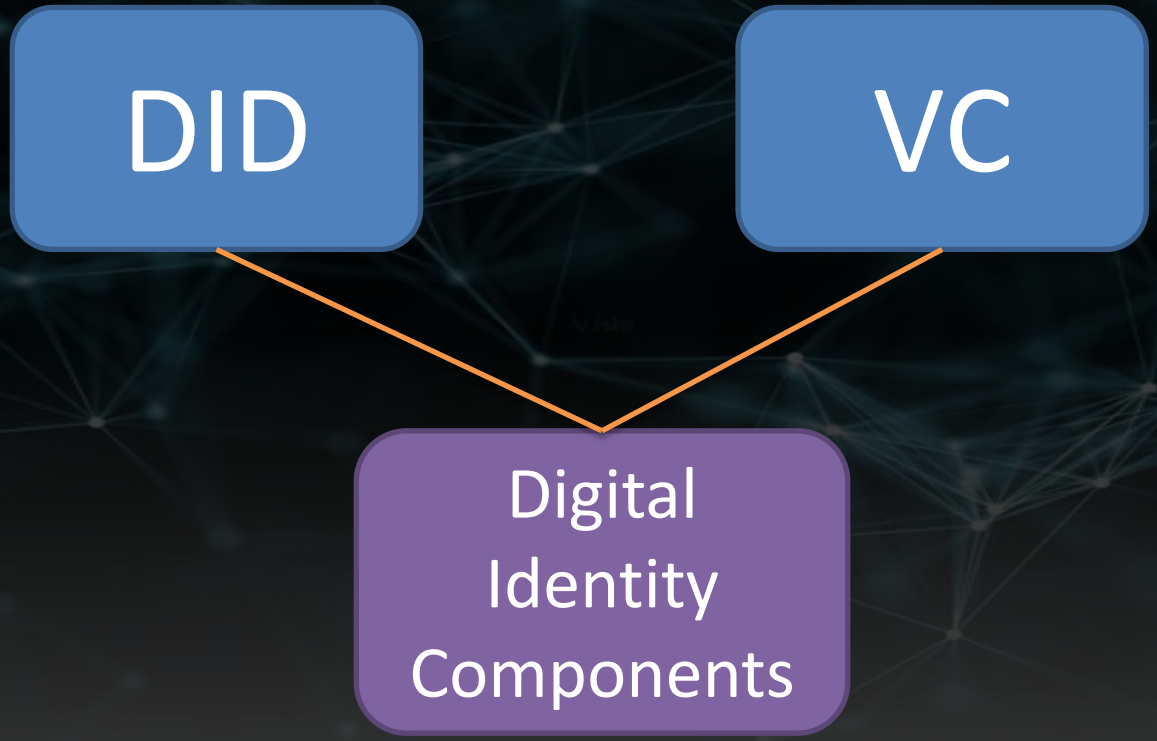

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DID

did:web:alice.com:port/path

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VC


```
{
    "@context": [
      "https://www.w3.org/2018/credentials/v1",
      "https://www.w3.org/2018/credentials/examples/v1"
    ],
    "id": "http://example.edu/credentials/1872",
    "type": ["VerifiableCredential", "AlumniCredential"],
    "issuer": "https://example.edu/issuers/565049",
    "validFrom": "2010-01-01T19:23:24Z",
    "credentialSubject": {
      "id": "did:example:ebfeb1f712ebc6f1c276e12ec21",
      "alumniOf": {
      "id": "did:example:c276e12ec21ebfeb1f712ebc6f1",
      "name": [
        {"value": "Example University", "lang": "en"},
        {"value": "Exemple d'Université", "lang": "fr"}
      ]
    }
}
```

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Format 1 : Natual Language

Format 2 : Structured format(e.g. : JSON,JSON-LD,Other structured formats)

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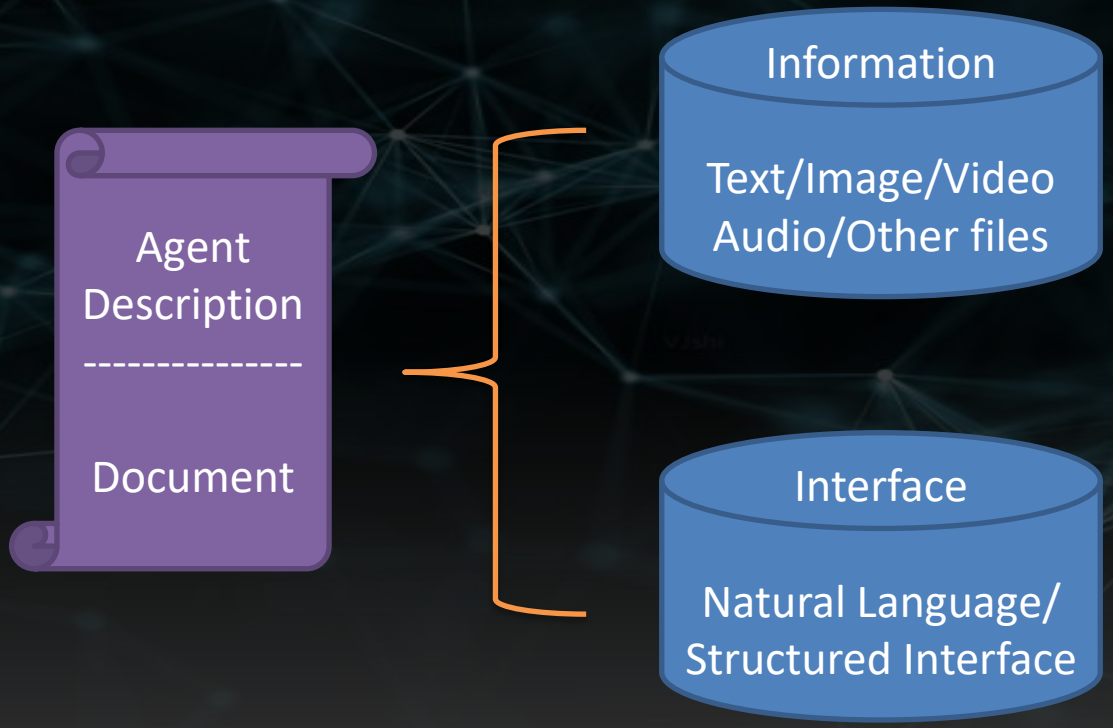

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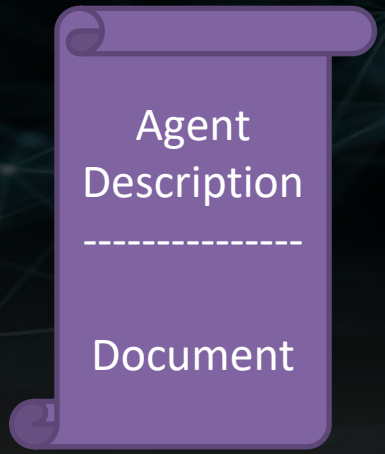

Security Mechanism : Mandatory
Include : Global Scope,Recource-Specific Scope,Precedence Rule

Integrity Verification : Mandatory
Proof : VC-compliant

...

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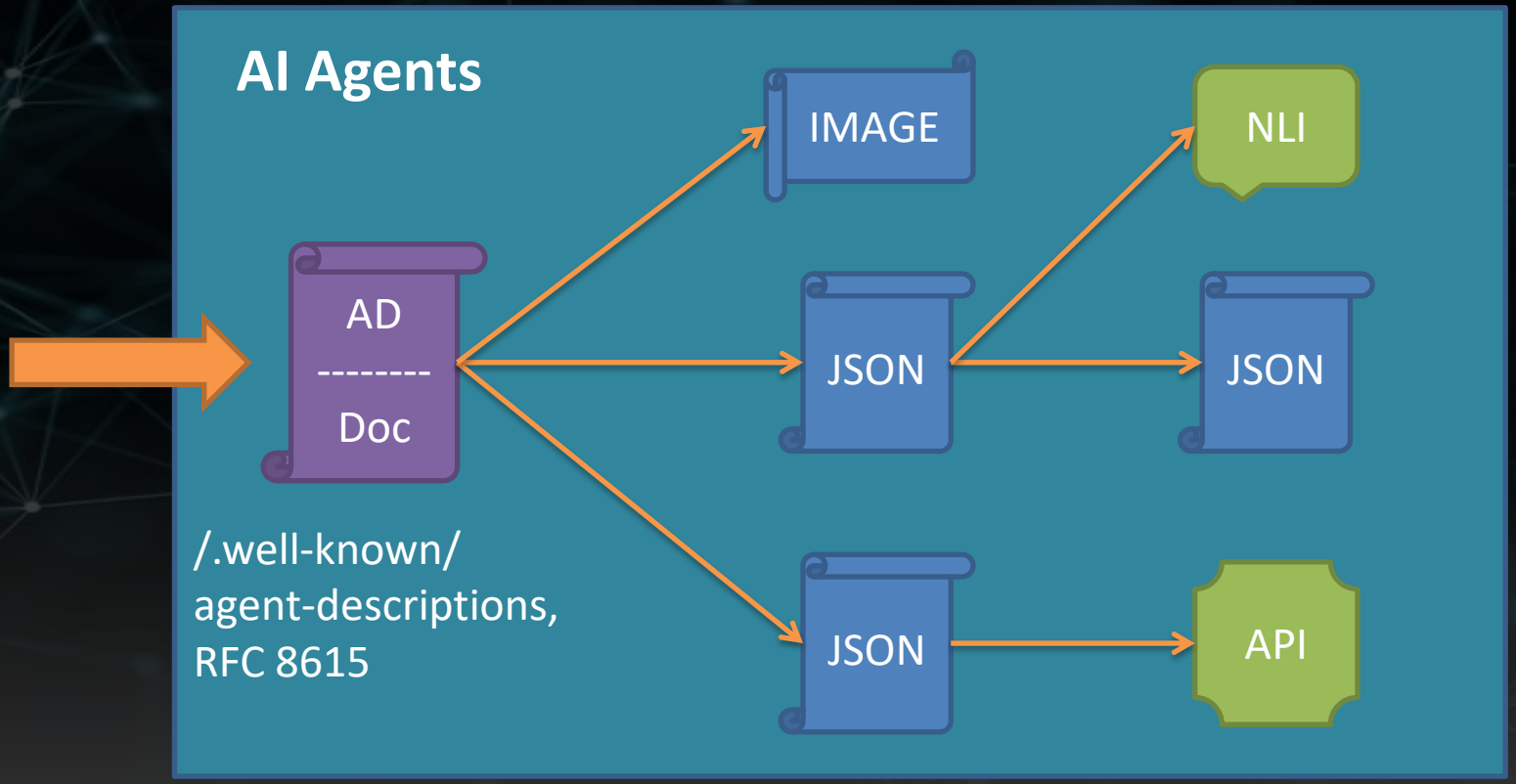

Registration : Self-Declaration Mode

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AI Agents

Name
Digital Identifier
Description
Address
Version
Capabilities
Services
Skills
Interfaces
Security Information

Register

Agent Registration Server

Registration : Centralized Registration Mode

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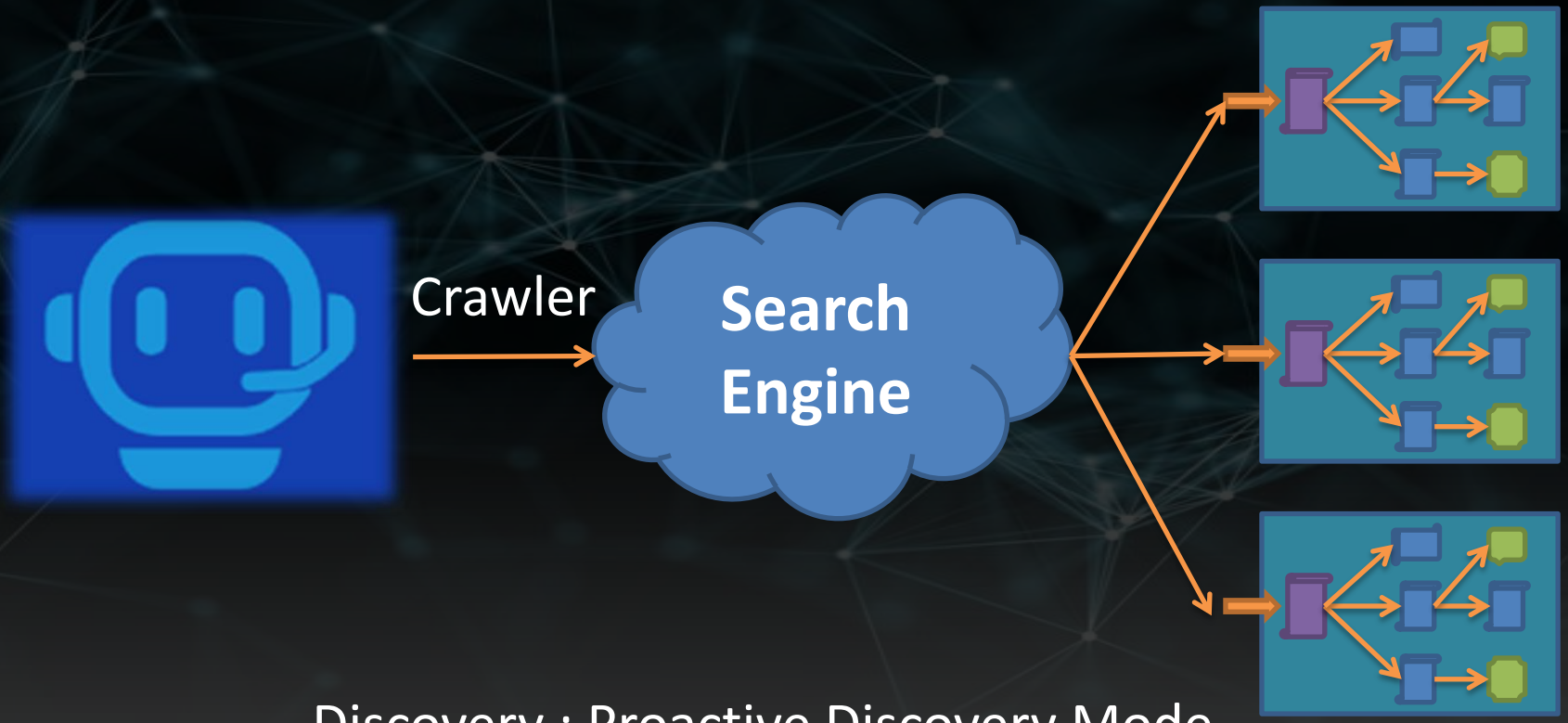

Discovery : Proactive Discovery Mode
(Corresponding to Self-Declaration Mode)

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Centralized Query Mode
(Corresponding to Centralized Registration Mode)

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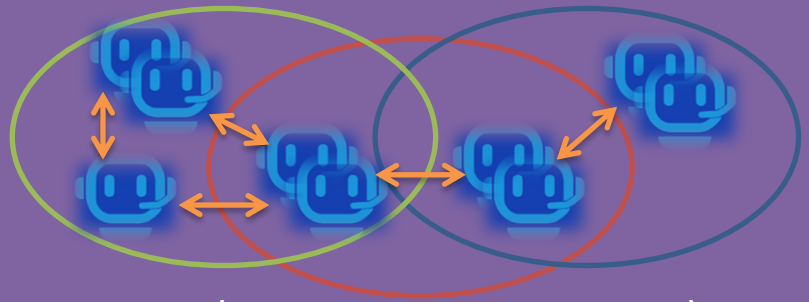
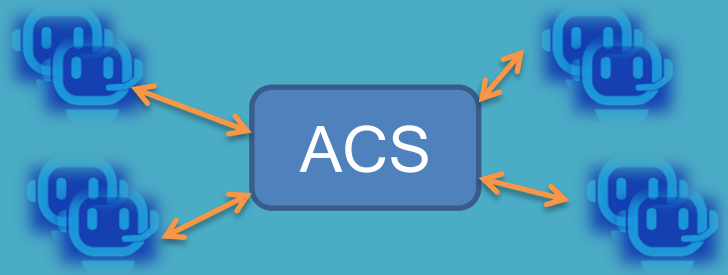
P2P (Direct Communication)

P2P (Indirect Communication)



Group (Direct Communication)



PUB/SUB (Indirect Communication)

Communication Mode (Message Mode)

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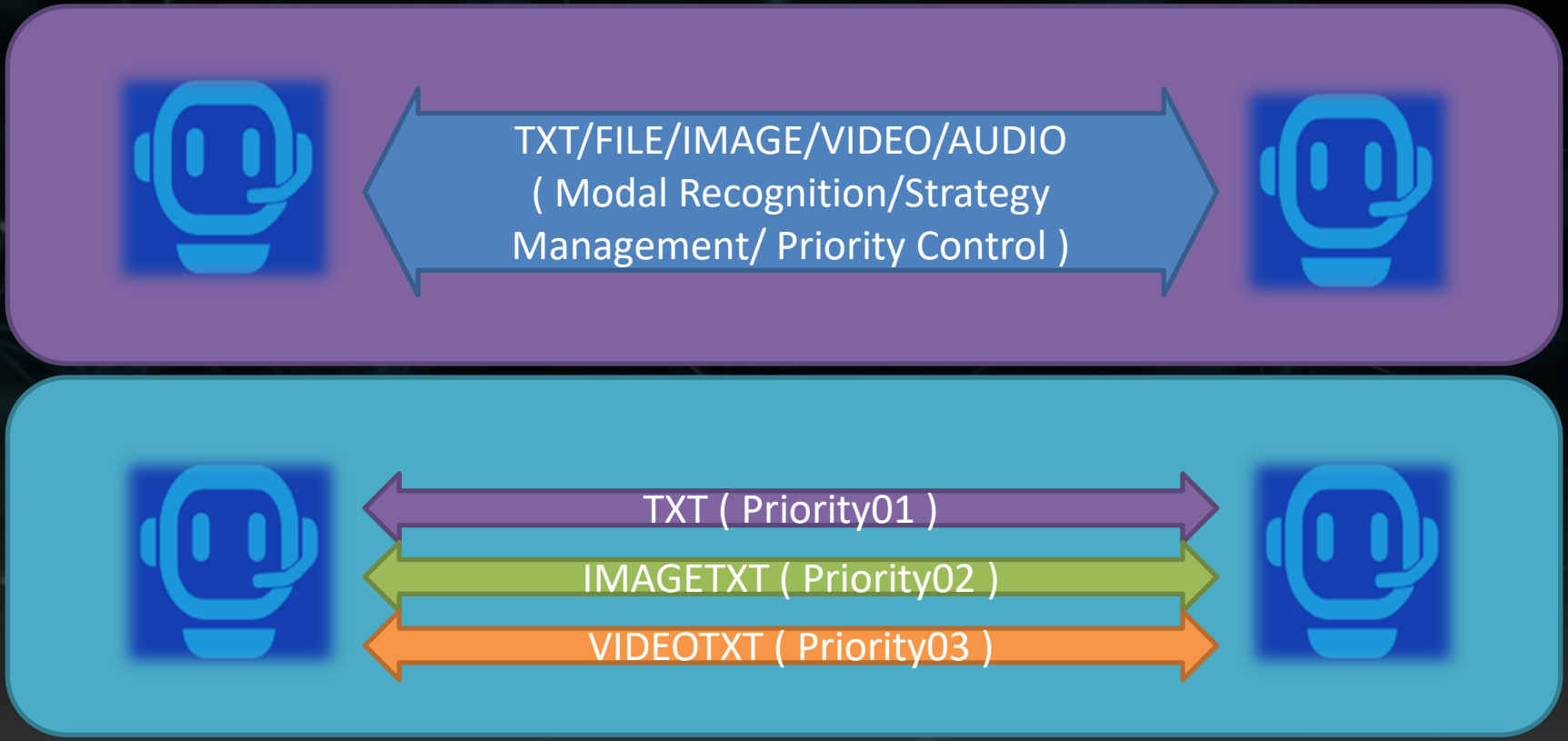

Multimodality

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AI Agent Application
(e.g. , Remote Care, Home Security, Smart Home, …)

AI Agent Comm Protocol Layer (ACPL)
(e.g. , JSON, JSON-LD, YAML, XML,…)

Application Layer (AL)
(e.g. , HTTP, SIP, …)

Transmission Layer (TL)
(e.g. , UDP, TCP, QUIC, …)

Protocol Stack Considerations

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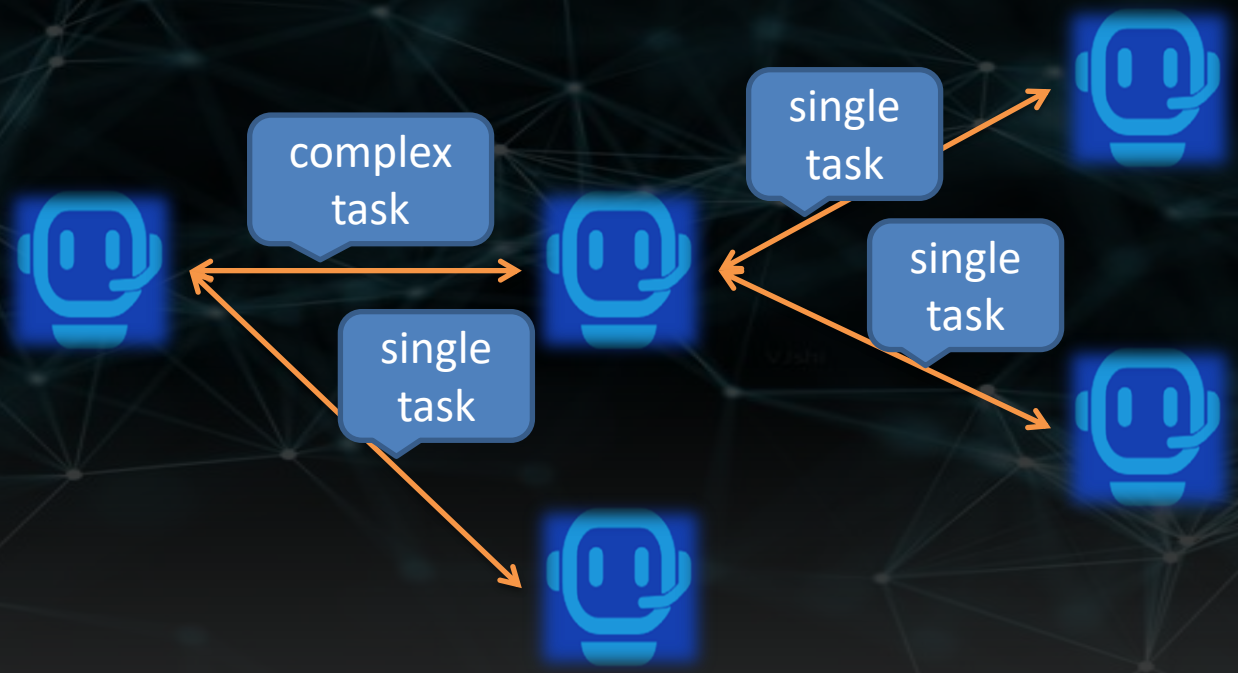

Task definition, planning and decomposition, status tracking, and result notification

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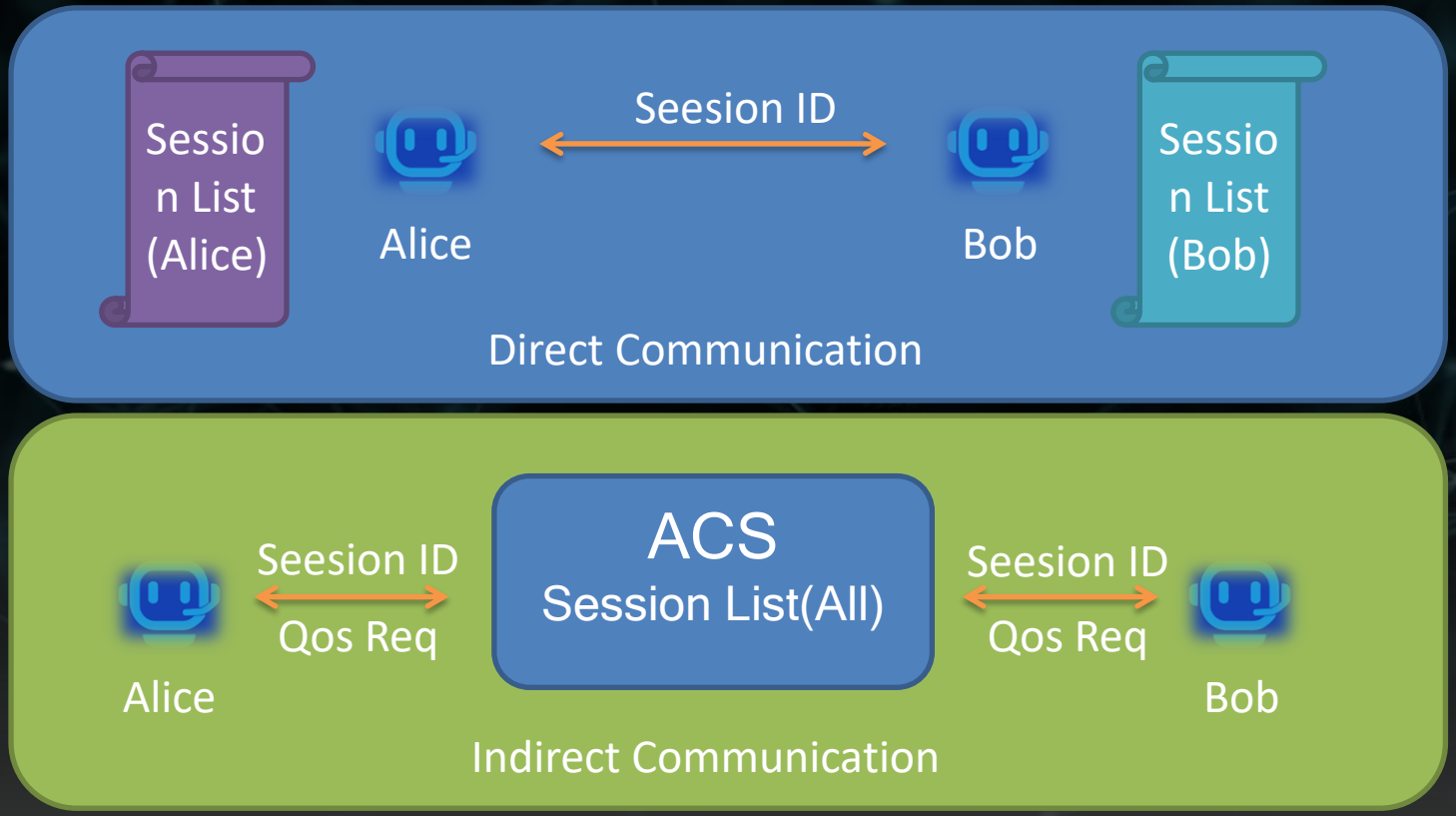

Session Management

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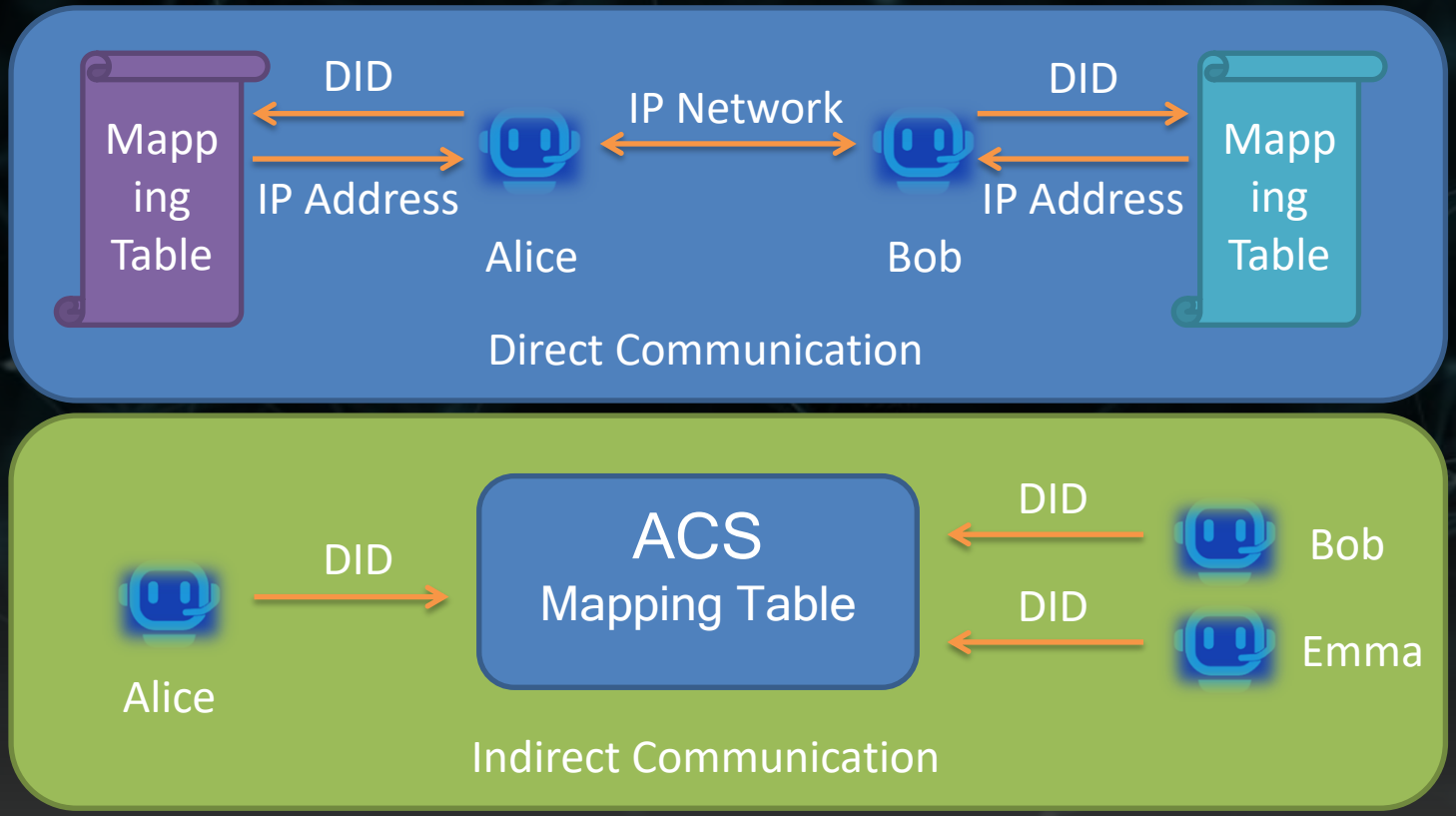

Routing I : ID-based

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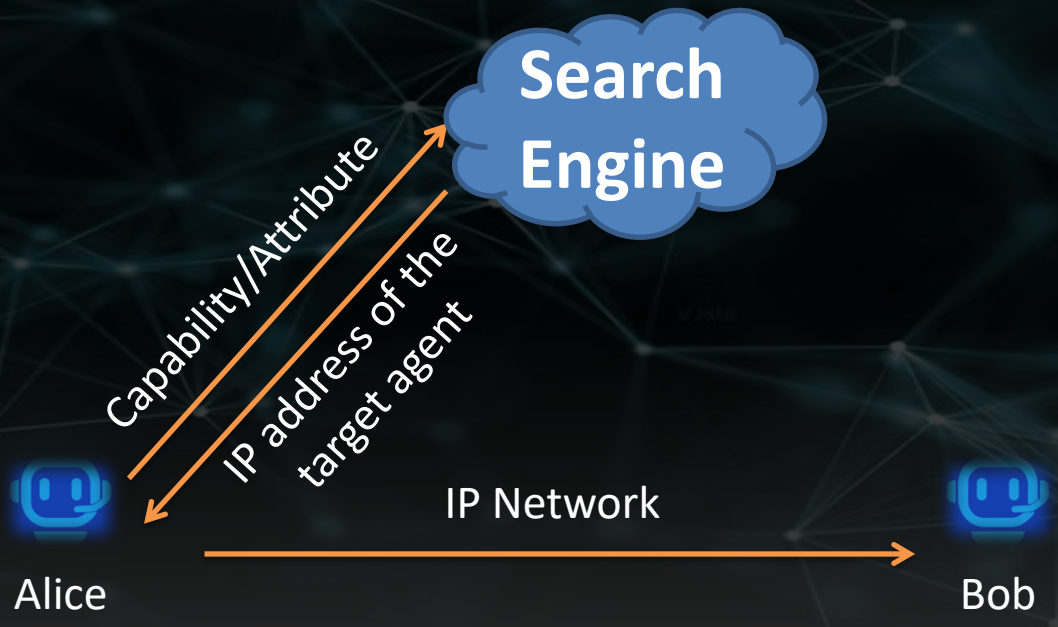

Routing II : Semantic-based

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Security Considerations

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Summary

- ✓ Introduce AI Agent Communication framework, including ARS and ACS.
- ✓ Identity: DID+VC.
- ✓ Agent Description: Natural language or Structured.
- ✓ Two registration and discovery mode: Self-Declaration and Centralized.
- ✓ Two communication mode: Direct mode and Indirect mode(through ACS).
- ✓ Multimodality support: txt, file, real-time audio/video
- ✓ Task management: task decomposition, status sync
- ✓ Two routing mode: ID-based and Semantic-based
- ✓ Protocol stack: Application layer, transport layer
- ✓ Security considerations

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Thanks !

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